

Edit-JEPA:
Latent World Models over Text Buffers
“Edit until perfect” — method, results, and open problems

TextJEPA project

July 9, 2026 (repo: `TextJEPA/`, runs: `runs/edit_*`)

Idea: the world is a mutable draft

- ▶ State = a **text buffer**: a draft solution to an iGSM-style problem (same generator as the discourse track; see companion deck for the problem sampler).
- ▶ Action = a **span edit** with an intent phrase that never leaks outcomes: delete step 3 . / insert green lamps at the start . / replace step 2 : recompute dark cups .
- ▶ Corruptions have **propagated consequences**: corrupting one step's value rewrites every downstream step *consistently* (change an assumption $\rightarrow$ later derivations change). One root cause $\Rightarrow$ several defects.
- ▶ Defect definition (symbolic, order-insensitive): a buffer step whose sentence $\neq$ the true step sentence, a missing necessary step, or an extra distractor step. **Solved** = zero defects.
- ▶ Generation = **edit until perfect**: latent search over candidate edits, execute best, re-encode, repeat.

Corruption model (exact) and a verbatim sample

From the perfect draft (necessary steps, topological order), sample independently: $k_{\text{wrong}} \sim \mathcal{U}\{0..2\}$ internal steps get a wrong stated value (propagated to descendants via the corrupted value map); $k_{\text{miss}} \sim \mathcal{U}\{0, 1\}$ necessary steps removed; $k_{\text{extra}} \sim \mathcal{U}\{0, 1\}$ distractor steps inserted; re-corrupt if defect count = 0. Repair trajectories: uniformly random fixing edit per step; with $p=0.1$ (0.25 in valgrad/anchor; max 1–2) a **vandal** edit (harmful distractor insert) $\Rightarrow$ the value head sees edits that *increase* distance. **Corrupted draft** (3 defects; step 3's wrong value propagates into step 4):

[1] so the number of large shells is 11 .

[2] so the number of silver cards is 17 .

[3] so the number of green books is 11 minus 17 = 5 . $\leftarrow$ wrong (true: 17)

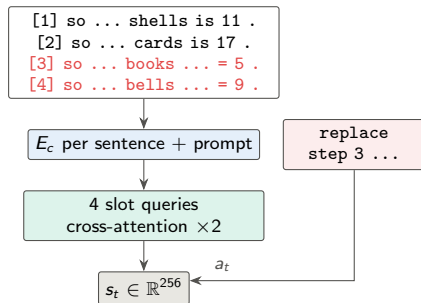
[4] so the number of golden bells is 5 times 17 = 9 . $\leftarrow$ consistent-but-wrong

missing: the step for green lamps

Fixing intents: replace step 3 : recompute green books . insert green lamps at the start .

Architecture: slot-based buffer encoder + shared JEPA core

- ▶ Chunk encoder E_c (identical to discourse track, 256-d).
- ▶ **SlotBufferEncoder**: 4 learned slot queries cross-attend (2 decoder layers, 4 heads) over [prompt chunks | buffer chunks (+ buffer position emb.)]; flatten slots $\rightarrow$ 256-d state. Length-invariant, order-aware.
- ▶ Action encoder, predictor F , LDAD ($n_{ops}=3$), macro hierarchy ($K=3$), value head: the *same* LatentDynamicsCore as the discourse track.
- ▶ Value target = **defects remaining** (distance to perfect).
- ▶ `edit_anchor`: predict the *frozen random-init* buffer embedding of the post-edit buffer from (s_t, a_t) — the collusion fix ported from the discourse track.



Objectives, training, planning

Losses: identical composite to the discourse deck (weights 1.0 pred / 1.0 rollout / 0.5 hierarchy / 1.0 VICReg / 2.0 Delta-JEPA / 1.0 value; anchor adds $\mathcal{L}_{\text{chunk}}$ w=2.0 on frozen buffer embeddings).

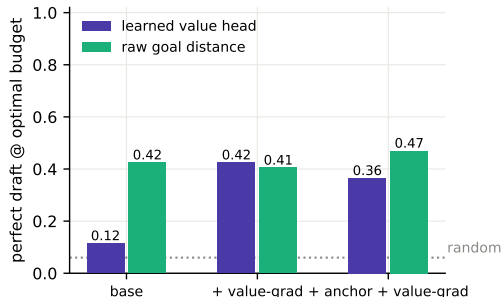
Labels: $\text{op} \in \{\text{delete, insert, replace}\}$; “was the edit a fix or vandalism” plays the goal-relevance role.

Training: batch 64 (each item encodes all $T+1$ buffer snapshots), 20 epochs $\times$ 60k fresh trajectories, otherwise identical protocol (AdamW 3×10^{-4} , EMA 0.99 $\rightarrow$ 0.999, RTX 6000, $\sim$ 70–90 min).

Planning: candidates from the edit interface = $\{\text{delete } p, \text{replace } p\}$ for every position + insert for every missing necessary step + 2 harmful distractor inserts (planner must *reject* them); score $V(F(s, a_i), s_{\text{init}})$; apply argmin in the environment; re-encode; budget = initial defects + slack. Success = **perfect draft within the optimal number of edits.**

Result — the geometry was never the problem

Claim: The buffer encoder's *raw* goal-distance geometry beats every learned edit value head — the published weakness was a value-head failure.



- ▶ Value heads: base 0.115 (random 0.06), valgrad 0.425, anchor 0.365 — all **slack-insensitive** (ties on no-op edits waste budget).
- ▶ Raw goal distance $\|\text{LN}(\hat{s}) - \text{LN}(s_{\text{goal}})\|_1$ (s_{goal} = encoded perfect buffer): base **0.425**, anchor **0.47** — 4× the base value head, no value supervision used at plan time.
- ▶ Geometry regularizers (straighten/monotonicity, cf. discourse deck) change nothing here: 0.445–0.485 — $\text{corr}(d^{\text{goal}}, \text{defects})$ was already ≈ 0.75 .

Probe summary (edit track)

probe (linear, val split)	base	valgrad	random-enc
edit op from Δs	1.00	1.00	1.00*
fix-vs-vandal from Δs	0.96	0.98	0.92
defects remaining from s_t	0.46	—	0.38
buffer length from s_t	0.91	—	0.73
claimed answer from s_t	0.23	—	0.28
true answer from terminal state	0.05	—	0.13

*intent phrases are trivially separable by op — the informative row is fix-vs-vandal. The two bottom rows show the **collusion signature** (trained < random control): outcome content is being smoothed away, as in the discourse track before its anchor fix. `edit_anchor` (frozen buffer anchor, $w=2.0$) trains $\mathcal{L}_{\text{chunk}}$ to 0.005 and lifts value-head planning to 0.365, but the linear *defect-count* probe stays flat — the goal signal lives in the *metric* (see previous slide), not in a linearly decodable counter.

Diagnosis & next steps

- ▶ **Why the base value head fails:** repairs are near-deterministic given the defect set, so state prediction is easy without a linear defect counter; but the *metric* structure (distance to the perfect buffer) is there from the start — use the goal-distance energy, or distill it into V .
- ▶ **Deployable goal states:** the oracle goal uses the true buffer; a deployable version needs a *goal description* encoder (“all steps correct”) or a self-predicted fixed point $F(s, a_{\text{stop}}) \approx s$.
- ▶ **Energy ties** are now the binding constraint (planner is slack-insensitive even at 0.47): ranking loss over the candidate set / advantage head $A(s, a)$ on $\Delta\text{defects}$.
- ▶ **Harder corruption:** multiple root causes, order violations, subtle wrong-parent corruptions (currently only wrong values / missing / extra).
- ▶ **Long-term:** draft-level macro edits (rewrite region), and the hierarchical planner shared with the discourse track.